

Tutorial IV – Mass Estimation

Aircraft weight has a direct impact on cruise drag and consequently, on mission fuel consumption. Furthermore, aircraft point performance depends on the actual mass. Therefore, early estimation of component weight is crucial for a successful design. Underestimating will ultimately lead to a degraded performance.



Basics

-  How is the take-off weight typically subdivided?
-  Name the three main aircraft groups included in operating empty weight.
-  Derive the algebraic relationship between aerodynamic drag, aircraft weight and C_L/C_D -ratio in horizontal flight.

Calculation tasks

Task 1-4 aim to determine the operating empty mass of the Airbus A320. For every task, compare the results to similar aircraft from the table on next page by building the respective fractions of gross weight.

HINT: Examine the table on next page. Note that units are in lbm. The expression GW stands for Gross Weight which can be considered to be equivalent to the maximum take-off mass $m_{TO,max}$.

- 1) Calculate the airframe (structure) mass $m_{airframe}$ of the Airbus A320 in lbm. It consists of:
 - Wings (use Torenbeek method)
 - Empennage
 - Fuselage
 - Nacelle & Pylons
 - Landing gear
- 2) Then also calculate the propulsion mass m_{prop} .
- 3) Determine the systems mass m_{sys} (by assuming the aircraft serves short-range routes).
- 4) Finally calculate the operating empty mass m_{OE} .
- 5) The certification specifications for civil aircraft postulate, that the aircraft structure must withstand vertical load factors up to $+3.75 g$. A novel flight control system now enables a limitation to $+3.0 g$ thanks to dynamic control surface deflections. Due to the reduced peak stress, the wings can therefore be built lighter (assuming the certification specifications change). How much do they weigh in % relation to the original wings?

Assumptions and Additional Data

Design dive speed: $V_D = 381 \text{ kn}$

Gravitational acceleration: $g = 9.81 \text{ m/s}^2$

Maximum vertical load factor: $n_{z,ult} = +3.75 g$

Assume that the aircraft serves short ranges, has a variable incidence stabilizer and that it is equipped with thrust reversal. The cabin is pressurized.

Type	Mc Donnell Douglas DC-9-30	MD-80	DC-10-10	Boeing 737-200	727-100	747-100	Airbus A300-B2
Flight Design Gross Weight, GW, lbs	108,000	140,000	430,000	115,500	160,000	710,000	302,000
Structure/GW	0.286	0.304	0.316	0.270	0.317	0.298	0.353
Power Plant/GW	0.076	0.079	0.077	0.071	0.078	0.062	0.076
Fixed Equipm't/GW	0.175	0.182	0.169	0.129	0.133	0.089	0.116
Σ Empty Weight/GW	0.538	0.564	0.561	0.521	0.552	0.498	0.559
Wing Group/GW	0.106	0.111	0.114	0.092	0.111	0.122	0.146
Empenn. Group/GW	0.026	0.024	0.032	0.024	0.026	0.017	0.020
Fuselage Group/GW	0.103	0.115	0.104	0.105	0.111	0.101	0.119
Nacelle Group/GW	0.013	0.015	0.020	0.012	0.024	0.014	0.023
Land. Gear Group/GW	0.039	0.038	0.046	0.038	0.045	0.044	0.045
Take-off Gross Wht, W_{TO} , lbs	108,000	140,000	430,000	115,500	160,000	710,000	302,000
Empty Weight, W_E , lbs	58,070	78,950	241,250	60,210	88,300	353,398	168,805










Roskam, J. (1985). Airplane design Vol. V, p. 153.

Equations

$$\tan(\Lambda_{m\%}) = \tan(\Lambda_{n\%}) - \frac{4}{AR} \cdot \left[\frac{m-n}{100} \cdot \frac{1-\lambda}{1+\lambda} \right]$$

$$m_W = 0.0017 \cdot m_{ZF} \cdot C_W \cdot b_S^{0.75} \cdot \left[1 + \sqrt{\frac{6.25}{b_S}} \right] \cdot n_{z,ult}^{0.55} \cdot \left(\frac{b_S}{t_{max} \frac{m_{ZF}}{S_{ref}}} \right)^{0.3}$$

with:

$$b_S = \frac{b}{\cos \Lambda_{50\%,W}}$$

$$C_W = 1 \cdot [x0.95 \ x0.95 \ x0.70]$$

- strut braced wings (y/n)
- undercarriage mounted elsewhere (y/n)
- engines mounted on wings (y/n)

$$m_T = C_T \cdot S_T \cdot \left(3.81 \cdot \frac{S_T^{0.2} \cdot V_D}{1000 \cdot \sqrt{\cos \Lambda_{50\%,T}}} - 0.287 \right)$$

with:

$$C_{HT} = 1 \cdot [\text{x1.10}]$$

→ Variable incidence stabilizer (y/n)

$$C_{VT} = 1 \cdot [\text{x1.15} \cdot \frac{S_{HT} \cdot z_{HT}}{S_{VT} \cdot b_{VT}}]$$

→ Horizontal tail mounted on vertical tail (y/n)

$$m_{fus} = 0.021 \cdot C_{fus} \cdot \sqrt{V_D \cdot \frac{d_{HT}}{w_{fus} + H_{fus}}} \cdot S_{fus}^{1.2} \text{ with } S_{fus} \approx l_{fus} \cdot \pi \cdot \left(\frac{w_{fus} + H_{fus}}{2} \right) \text{ \& } l_{HT} \approx x_{0,HT} - x_{0,W}$$

with:

$$C_{fus} = 1 \cdot [\text{x1.04} \text{ x1.07} \text{ x1.08}]$$

→ pressurized cabin (y/n)
→ main landing gear attached to fuselage (y/n)
→ rear-fuselage mounted engines (y/n)

$$m_{nac} = 0.065 \cdot T_{TO}$$

$$m_{LG} = C_{LG} \cdot (C_1 + C_2 \cdot m_{TO,max}^{0.75} + C_3 \cdot m_{TO,max} + C_4 \cdot m_{TO,max}^{1.5})$$

with:

	C_1	C_2	C_3	C_4
Nose gear	20	0.10	0	$2 \cdot 10^{-6}$
Main gear	40	0.16	0.019	$1.5 \cdot 10^{-5}$

$$C_{LG} = 1 \cdot [\text{x1.08}]$$

→ High-wing configuration (y/n)

$$m_{prop} = 1.15 \cdot C_{prop} \cdot n_{eng} \cdot m_{eng}$$

with:

$$C_{prop} = 1 \cdot [\text{x1.18}]$$

→ Thrust reversal (y/n)

$$m_{sys} \approx C_{sys} \cdot m_{TO,max}$$

with:

$$C_{sys}$$

→ =14% for short-range transport aircraft
→ =11% for medium-range transport aircraft
→ = 8% for long-range transport aircraft